PIYUSH GOEL

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PROFESSIONAL SUMMARY

AI Engineer and Backend Developer specializing in Agentic RAG architectures, Multimodal VLMs, and production LLM orchestration. Proven track record of building low-latency document intelligence pipelines and scalable microservices using Python, FastAPI, LangGraph, Qdrant, and Java (Spring Boot) to bridge Generative AI with high-concurrency backend systems.

SKILLS

AI & LLM Engineering: Generative AI, Agentic & Corrective RAG (CRAG), LangGraph, LangChain, Multimodal VLMs (Qwen-VL), LLMs (GPT, Gemini, Groq, Ollama), Embeddings (BGE-M3), Cross-Encoder Reranking, Prompt Engineering
Programming & Backend: Python, Java, SQL, FastAPI, Spring Boot, REST APIs, High-Concurrency Architectures, JWT
Vector DBs & Tools: Qdrant, ChromaDB, Docling, RapidOCR, PyMuPDF, MySQL, MinIO, Docker, Git, React, Agile

PROJECTS

NewsLens-AI | Multimodal Document Intelligence & Agentic RAG Platform 2026
Technology Used: *Python, LangGraph, Qdrant, Docling, Qwen-VL, FastAPI, MySQL, React, MinIO, BGE-M3*

- Architected a multimodal document intelligence pipeline using Docling and RapidOCR to parse, segment, and index high-density broadsheets into Qdrant vector databases while preserving complex 2D reading orders.
- Built a high-throughput visual pipeline with Qwen-VL to extract trapped financial charts into structured Markdown, cutting VLM inference latency from 48s to <3s per asset via patch token downsampling.
- Designed a Corrective RAG (CRAG) LangGraph agent featuring cross-encoder reranking and a 12-domain probabilistic classifier to ensure high-precision, hallucination-free retrieval.

DocuMind RAG | Multi-Model Contextual Generative AI Application 2026
Technology Used: *Python, FastAPI, ChromaDB, Gemini, Groq, Ollama, PyMuPDF, python-docx*

- Architected an end-to-end RAG pipeline to perform grounded, citation-backed querying over unstructured enterprise documents, optimizing semantic chunking for retrieval accuracy and low latency.
- Engineered modular data ingestion, cleaning, and tokenization workflows (PyMuPDF, python-docx) integrated with dense vector embeddings persisted in ChromaDB for high-recall similarity search.
- Built a fault-tolerant LLM orchestration layer to dynamically route inference across Gemini, Groq, and local Ollama models, minimizing cloud token overhead and API fallback latency.

Campus Connect | Scalable Production Backend Jan 2026 – Present
Technology Used: *Java, Spring Boot, MySQL, RESTful APIs, JWT, Git*

- Engineered a high-concurrency backend infrastructure supporting production-ready REST APIs, real-time data workflows, and stateless JWT authentication for 250+ active users.
- Optimized MySQL query execution plans, relational indexing, and backend pipelines to maintain low-latency API responses and high system observability under concurrent load.

EXPERIENCE

Infosys Delhi, India
Java Technology Stack Intern *May 2024 – July 2024*

- Developed and integrated scalable backend modules using Java, Spring Boot, and SQL within an enterprise application architecture, streamlining data processing workflows across cross-functional sprints.
- Executed query optimizations, unit testing, and peer code reviews within Agile workflows to ensure production-grade reliability and adherence to software engineering standards.

EDUCATION

BVICAM, GGSIPU Delhi, India
Master of Computer Applications (MCA) | CGPA: 8.5 *2025 – 2027*

IIT Madras Remote
BS in Data Science and Applications | CGPA: 8.0 *2025 – Ongoing (Flexible Track)*

IITM, GGSIPU Delhi, India
Bachelor of Computer Applications (BCA) | CGPA: 8.7 *2021 – 2024*

CERTIFICATIONS & ACTIVITIES

Leadership & Recognition: Public Relations Head, IEEE BVICAM | NPTEL Star Learner (IIT Kanpur)
Languages & Interests: English (Professional), Hindi (Native) | Badminton, Geopolitics & Current Affairs